

State-Selected Reheating in Exact Z_6 Chronometry

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

33. Why hard asymmetric reheating is dangerous

The cyclic cancellation of lower vacuum harmonics relies on exact microscopic symmetry. A permanent sector-specific reheating coupling is a hard Z_6 -breaking spurion and generically feeds lower harmonics back into the vacuum potential. The correct architecture is therefore

exact Z_6 action + temporarily asymmetric state.

This distinction is strongly supported by later cyclic finite-density models [Delaunay2026].

34. Orbit-symmetric reheaton and transient selector

Introduce two complete reheaton orbits X_k, Y_k with oriented cyclic mixing. Light eigenstates have the schematic form

$$R_k = \cos \theta X_k + \sin \theta Y_{k-1}.$$

A selected population of R_0 decays into sectors 0 and 5. A complete selector orbit Q_k has one-hot inflationary branches. On the Q_0 branch, all reheaton copies except R_0 are made kinematically inaccessible by an exactly Z_6 -invariant mass pattern.

The key chronology is

$$\Gamma_{\phi \rightarrow R_0 R_0} \gg \Gamma_Q \gg \Gamma_R.$$

The inflaton first produces the labelled state. The selector returns to the symmetric origin before the reheaton decays. The asymmetry survives in occupation numbers, not in late-time masses or couplings.

At the first homogeneous level, a branching ratio $\Gamma_5/\Gamma_0 = 1/256$ gives $T_5/T_0 = 1/4$. The nonlinear cascade later corrects the exact mixing required to achieve that final temperature ratio.

35. Perturbations and isocurvature

For fixed branching fractions,

$$\rho_i = B_i \rho_R,$$

and radiation curvature perturbations satisfy

$$\zeta_i = \zeta_R + \frac{1}{4} \delta \ln B_i.$$

If the branching fractions are fixed constants,

$$\delta B_0 = \delta B_5 = 0,$$

so

$$S_{50} = 3(\zeta_5 - \zeta_0) = 0.$$

The extra radiation is adiabatic at linear order. This would fail if the mixing angle were a light modulus, because its fluctuation would be amplified by the small adjacent branching fraction. The benchmark keeps the mixing and selector sectors heavy during inflation.

The initial chronometric phase fluctuation

$$\delta x_{\text{reh}} = \frac{H_*}{2\pi f_a}$$

is strongly damped by threshold, QCD, and baryon focusing. The staged calculation found a transfer magnitude of about 10^{-6} by recombination and a residual QCD-lock fluctuation near 10^{-23} . This remains a reduced perturbation calculation, not a full Einstein-Boltzmann likelihood.